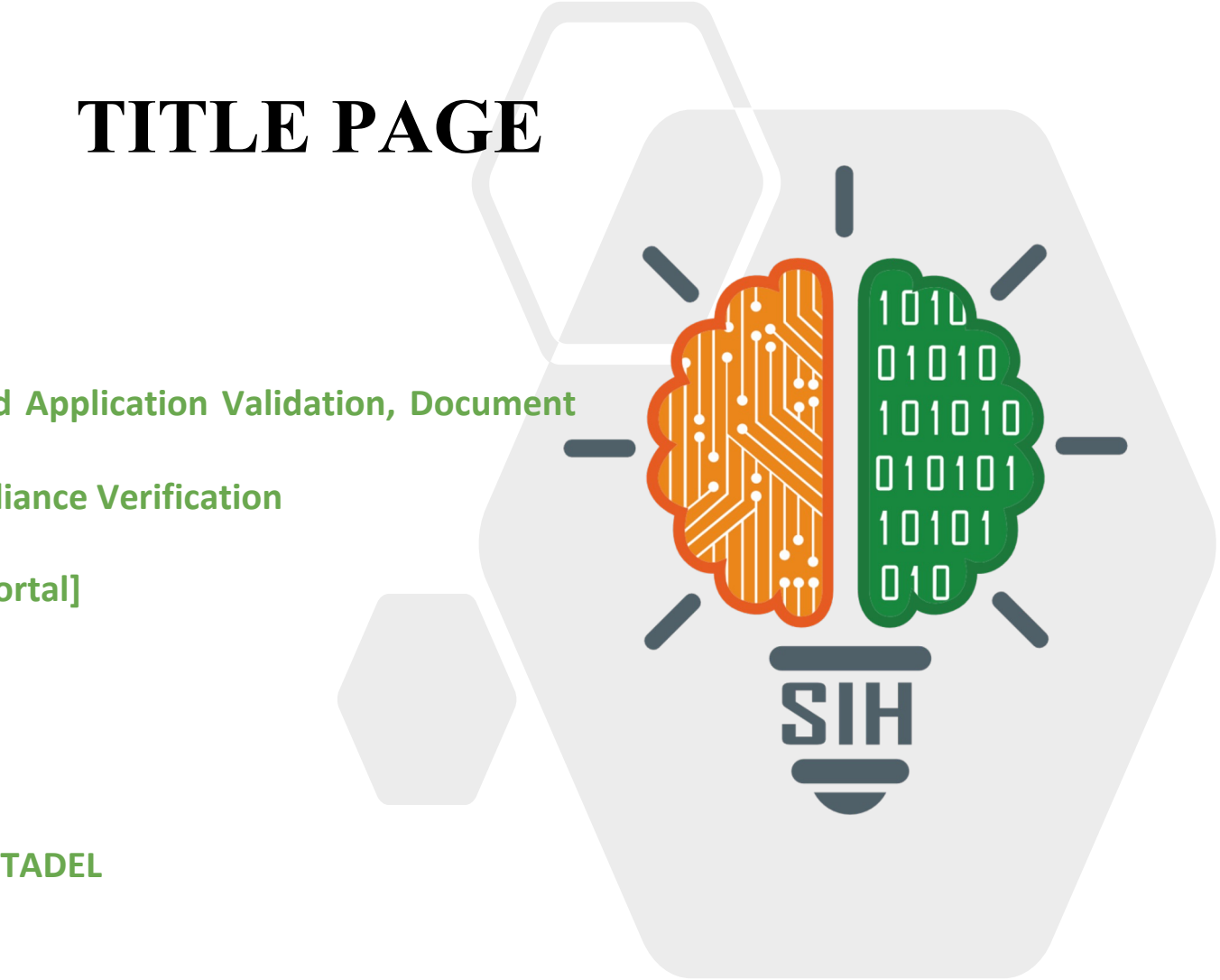
SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID:** 26261
- **Problem Statement Title:** AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
- **Theme:** [Theme as listed on the SIH portal]
- **PS Category:** Software
- **Team ID:** BMS/SIH2026/54
- **Team Name (Registered on portal):** CITADEL



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IDEA TITLE

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Proposed Solution: DossierGraph is an on-premise AI pre-scrutiny engine that validates an institution's whole AICTE approval dossier: every document against its **issuing authority**, against **20 APH 2024-27 rules**, and against **every other document**. Officers get evidence-linked findings and an inspection list; they make every decision.

Addressing the Problem

Self-disclosure is trusted at entry

EoA letters list deficiencies "based on Self Disclosure", fixed up to 6 months later.

→ Every document cross-checked at submission

Fraud is caught late, if at all

APH Cl. 2.3.5(b): a fraudulent document means a criminal case, yet detection rests on random EVC visits.

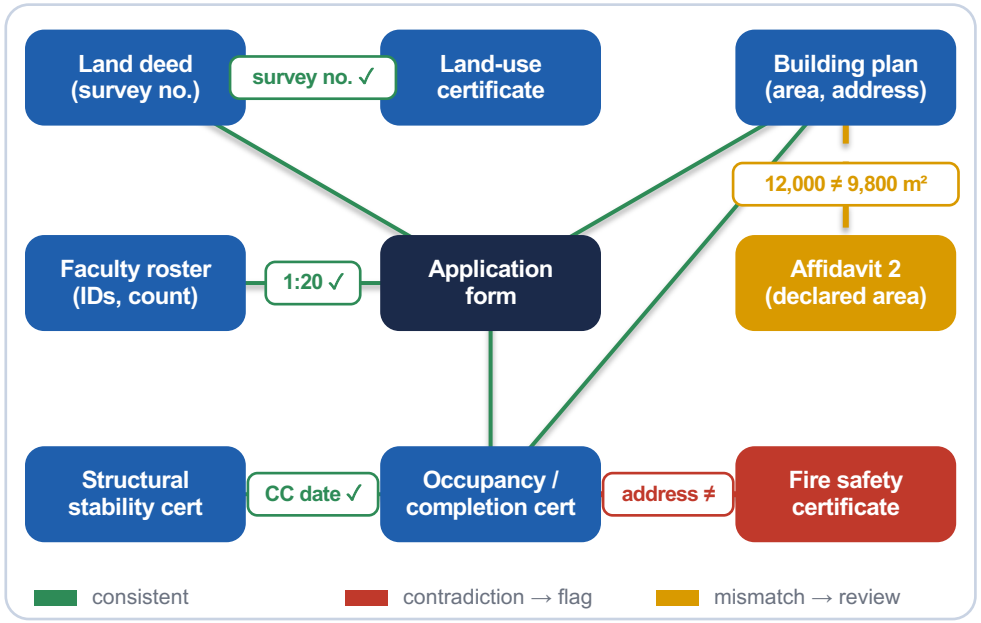
→ Forgery indicators + issuer checks before scrutiny

Officers re-check the same papers

Deeds, NOCs, OCs and faculty rosters are re-verified by hand every cycle.

→ Severity-ranked queue, clause + evidence per flag

Core idea: the dossier cross-consistency graph



Innovation & uniqueness

Whole-dossier checks
not single-document KYC

Measured accuracy
on an open, labelled benchmark

Rules as code
AI advises, officer decides

Solution Overview

Verify first

Digital signatures (DSC), QR codes and DigiLocker records checked at the issuing source before any forensics.

Dossier graph + APH rules

Facts linked across documents; 20 versioned rules citing APH clauses (land, building, fire, faculty).

Forensics only where needed

Unsigned scans get PDF-revision, metadata, ELA and photo-reuse checks, shown as advisory indicators.

Officer workstation + trust layer

Evidence viewer, 6 verification statuses, 2-officer rule for adverse outcomes, tamper-evident log.

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TECHNICAL APPROACH

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One dossier, end to end	Intake 1	Extract 2	Verify 3	Cross-check 4	Forensics 5	Decide 6
What the officer sees	Drag-drop dossier; checklist ticks as each document is recognised	Extracted fields highlighted on the page; unsure OCR fields editable	Status chip per document: verified, mismatch, expired, pending…	Graph view: red edges mark contradictions between documents	Heat-map on suspicious regions, labelled "indicator"	Severity-ranked queue: accept, override with reason, escalate
What the system does, and its call	Virus scan, PDF disarm & rebuild, per-document encryption `intake.scan() → ClamAV`	Classify 8 doc types; OCR; extract facts with page coordinates `extract() → PaddleOCR`	Validate DSC / QR; fetch the issuer's record and compare `verify() → pyHanko`	Link facts across documents; evaluate the APH rule pack `rules.eval() → aph.yaml`	PDF revisions + metadata, ELA, perceptual-hash photo reuse `forensics() → OpenCV`	Write finding + evidence crop; append to hash-chained log `audit.append() → PG`
Model or store	**MinIO + Vault** encrypted object store, per-doc keys	**PaddleOCR** English + Hindi, runs on-prem	**pyHanko + adapters** DSC, QR, DigiLocker; mocks where no API	**Rule engine + PG** versioned YAML rules, 20 APH clauses	**OpenCV + pikepdf** ELA, PDF structure, imagehash	**Next.js workstation** Keycloak MFA + OPA access rules
Output, and what's guaranteed	**dossier (encrypted)** never leaves AICTE servers	**facts.json** every value linked to its page	**status per document** issuer data beats forensics	**findings + APH clause** rule version frozen at submission	**advisory indicators** never auto-rejects	**signed report + EVC list** 2nd officer for adverse outcomes

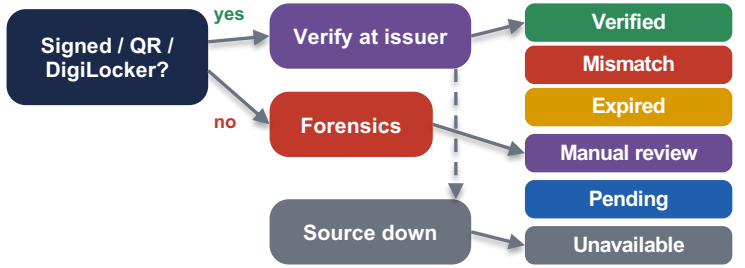
Rules as code: a real APH 2024-27 clause

```
rule_id: BLDG_FIRE
aph_ref: "Cl. 6.17(i), Annexure-14"
applies_to: [EoA, NEW_INSTITUTION]
check: fire_cert.valid_until >= ay_start
    and fire_cert.address ~ plan.address
severity: high
on_fail: "Upload valid Fire Safety Cert"
```

Technology stack (all on-prem)

Next.js · Tailwind · FastAPI · Python · Celery · Redis · PostgreSQL · MinIO · PaddleOCR · OpenCV · Keycloak · Vault · Docker · Kubernetes · pyHanko · ClamAV

Verification decision logic



Trust layer: AES-256 encryption per document · Keycloak MFA + OPA role/region rules · hash-chained, tamper-evident audit log · HMAC-hashed faculty IDs · on-prem models, no third-party AI APIs

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FEASIBILITY AND VIABILITY

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Feasibility & Viability

Technical
- **Mature open-source parts:** PaddleOCR, pyHanko, OpenCV; no custom model training needed
- **Verify-first:** signed / DigiLocker documents skip forensics, cutting false flags

Operational
- **Plugs in before the Scrutiny Committee:** Scrutiny → EVC → SHC → Executive Committee chain unchanged
- **One engine, two sides:** institutions run the same checks as a pre-check before submitting

Economic
- **On-prem, no per-document API fees:** scale by adding worker nodes on government servers
- **Targeted inspections:** EVC visits focus on unresolved items, not full re-checks

Legal & Privacy
- **DPDP Act 2023 + Rules 2025:** encryption, access logs, HMAC-hashed IDs, retention limits
- **MeitY AI Governance Guidelines (2025):** human oversight; officer makes every decision

Proof of accuracy
- **Open synthetic benchmark:** 200 institutions · ~2,000 documents · 8 labelled tamper types
- **We report:** precision / recall per tamper type and false flags per genuine application

POTENTIAL CHALLENGES & STRATEGIES TO OVERCOME THEM

POTENTIAL CHALLENGES & RISKS	MITIGATION STRATEGIES
Poor or regional-language scans: faded stamps, Hindi and state-language NOCs	**Indic OCR + per-field confidence:** low confidence goes to manual review, never auto-fails
Sophisticated forgeries: edits that pass image forensics	**Issuer check first:** multi-signal forensics (PDF structure + ELA + reuse), always advisory
False flags overload officers: noise erodes trust in the tool	**Benchmark-tuned thresholds:** false flags per genuine application tracked; severity-ranked queue
No public APIs: fire NOCs, land records, AICTE faculty registry	**Pluggable adapters:** spec-faithful mocks now, 'Source unavailable' + retry; MoUs on the roadmap
Rules change: APH revisions; the VBSA Bill may replace AICTE norms	**Versioned rule packs:** clause IDs per rule; dry-run on past applications before publishing

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IMPACT AND BENEFITS

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Where DossierGraph plugs into AICTE's approval chain (APH 2024-27)

Institution applies — NSWS / AICTE portal → **DossierGraph** — AI pre-scrutiny → **Scrutiny Committee** — zero-deficiency check → **Expert Visit Committee** — offline or online → **Standing Hearing Cttee** — show-cause replies → **Executive Committee** — LoA / LoR → **Standing Appellate Cttee** — appeal within 7 days

▲ flags + evidence to Scrutiny · 'verify on site' list to the EVC

Potential Impact on Target Audience

AICTE scrutiny officers

Severity-ranked queue; every flag shows the APH clause and the evidence crop

Institutions

Pre-check before submission: fix deficiencies first, fewer re-scrutiny rounds

Expert Visit Committees

A 'verify on site' list of only what could not be confirmed remotely

Students and parents

Approved institutions' land, building, safety and faculty claims checked

Benefits of the Solution

Social
Students protected from institutions with unverified infrastructure or faculty claims

Economic
Officer and inspection hours spent on real issues, not routine re-checks

Operational
Deficiencies caught before scrutiny; fewer back-and-forth cycles

Governance
Same versioned rules for every application; tamper-evident trail for appeals and RTI

Environmental
Digital-first verification; fewer physical re-inspection trips

Today vs. with DossierGraph

Today	With DossierGraph
Documents trusted on self-disclosure	Cross-checked against issuers and each other
Deficiencies fixed up to 6 months after EoA	Fixed before submission via pre-check
Fraud found by random EVC visits	Every dossier screened; indicators with evidence
Findings vary by reviewer	Same APH rule version for every application
Inspectors re-check everything	EVC targets only unresolved items

KPIs we will measure and report

- **Precision / recall per tamper type**
- **False flags per genuine application**
- **Minutes per application vs. manual**
- **% documents verified at source**

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RESEARCH AND REFERENCES

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Regulatory: AICTE

• AICTE Approval Process Handbook 2024-25 to 2026-27 (Cl. 2.3.5, 2.4, 6.14, 6.17, 6.18, 7.4; Annexures 3, 5, 14) https://cdnbbsr.s3waas.gov.in/s35938b4d054136e5d59ada6ec9c295d7a/uploads/2025…

• AICTE Grant of Approval guidelines 2024-27 (EVC, SHC, SAC process; fraudulent documents) https://indiankanoon.org/doc/163708613/

• Sample AICTE EoA 2026-27 letter listing deficiencies based on self-disclosure (VESIT) https://vesit.ves.ac.in/storage/aicte/1775193851EOA%20Report%202026-27%20(2).…

Data protection and AI governance

• Digital Personal Data Protection Act 2023 & DPDP Rules 2025 (notified Nov 2025) https://ssrana.in/articles/meity-notifies-final-digital-personal-data-protect…

• MeitY India AI Governance Guidelines (Nov 2025): human oversight, accountability https://en.vikaspedia.in/viewcontent/e-governance/digital-india/india-ai-gove…

Verification infrastructure

• DigiLocker Requester Entity Locker API Specification, NeGD (28-11-2024) https://entity.digilocker.gov.in/assets/img/Requester%20-%20Entity%20Locker%2…

• AICTE circular on NAD / DigiLocker adoption (06/2021) https://cdn.digilocker.gov.in/nad/assets/circulars/4%20Circular_AICTE_NAD_Dig…

• pyHanko: PDF digital-signature validation https://pyhanko.readthedocs.io/

Research and prior work

• Grobler et al., A Technique for the Detection of PDF Tampering or Forgery, arXiv:2507.00827 (2025) https://arxiv.org/abs/2507.00827

• PaddleOCR: multilingual OCR toolkit (PaddlePaddle) https://github.com/PaddlePaddle/PaddleOCR

• SIH 2024 AICTE problem statements SIH1729 (AI approval portal) and SIH1730 (AI inspection) https://www.driems.ac.in/wp-content/uploads/2024/08/SIH_2024_PS.pdf

How we compare

Capability	KYC tools	Typical SIH entry	DossierGraph
Verify at issuer	✓	Partial	✓
Whole-dossier cross-checks	✗	Partial	✓
APH rules with clauses	✗	Partial	✓
Measured accuracy, open data	Partial	✗	✓
On-prem, DPDP-aligned	Partial	Partial	✓

KYC tools: e.g. IDfy, HyperVerge (tamper checks for KYC and lending). Typical SIH entry: public SIH26261 repositories, Oct 2026.

Regulatory facts we build on

Cl. 2.3.5(b)
Fraudulent document → criminal case against Principal and Trust

6 months
Window to fix self-disclosure deficiencies in EoA letters

Nov 2025
DPDP Rules notified; MeitY AI guidelines issued

30 yrs / 5 yrs
Min. land lease (Cl. 6.17a); stability cert validity (6.17h)